



CADF: Real-time multi-biosignal modal recognition with causality-aware dimension fusion

Jing Tao ^a, Zhuang Li ^a, Lin Wang ^b, Dahua Shou ^{a,*}

^a Future Intelligent Wear Centre, School of Fashion and Textiles, The Hong Kong Polytechnic University, Hung Hom, Kowloon 999077, Hong Kong, China

^b Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518000, China

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ABSTRACT

Real-time analysis of multiple biological signals offers social media systems valuable insights into user engagement, but capturing the complex temporal dynamics and inter-signal relationships remains a challenge. This study introduces a novel framework, CADF (Causality-Aware Dimension Fusion), for real-time multi-biosignal modality recognition. CADF introduces a causality-aware temporal encoder that preserves temporal causality while effectively modeling long-term dependencies in one-dimensional signals. Additionally, the time series data is converted to extract 2D spatial masks. The bi-dimensional features are fused to identify modalities with the aid of a streamlined MultiHead mechanism. Extensive experiments on the DSADS, WESAD, and CAP datasets show that CADF reduces the number of parameters by at least 58% and improves the accuracy by 8% compared to the SOTA model. In particular, the accuracy of the three-classification emotion recognition task reached 95%. These results emphasize the effectiveness and efficiency of CADF in real-time biosignal analysis, with important implications for user-centric applications.

1. Introduction

The rapid advancement of smart wearable technologies (Cooper & Armentano, 2022) has revolutionized the way physiological signals are collected and analyzed. Wearable devices, sense an individual's physical and emotional state in real time by capturing large amounts of biosignals, have enabled a wide range of applications in health monitoring (Kim, Shin, & Yoon, 2017), emotion recognition (Li et al., 2024), and human-computer interaction (Wang & Zhang, 2022). Among these applications, one particularly promising area is the integration of physiological signal analysis into social media platforms to enhance personalized recommendations (Liu, Xu, Zhang, & Jiang, 2023). By leveraging real-time physiological responses, social media systems can better understand user engagement, emotional states, and preferences, thereby improving content recommendation accuracy and user experience (He, Liu, & Jung, 2024; Liu, Xu, & Zhou, 2024).

Multivariate and time-dependent are two prominent features of this field (Yu et al., 2024). Since multiple sensors are used to capture different types of physiological signals, each modality's data provides different aspects of information that must be effectively combined to gain a comprehensive understanding of an individual's state (Wang, Zhang, Chen, & Cao, 2025). These multiple variates

* Correspondence to: QT719, The Hong Kong Polytechnic University, 11 Yuk Choi Rd, Hung Hom, Hong Kong.

E-mail address: dahua.shou@polyu.edu.hk (D. Shou).

¹ Prof. Shou earned his Ph.D. from PolyU and then worked as a Postdoctoral Fellow consecutively at The University of Sydney and Cornell University. Prof. Dahua Shou is Limin Endowed Young Scholar in Advanced Textiles Technologies in The Hong Kong Polytechnic University (PolyU), and an Associate Professor at the School of Fashion and Textiles (SFT).

are not only heterogeneous but also highly time-correlated (Shan et al., 2024), with each signal evolving over time in a dynamic and interdependent manner. Because changes in physiological states tend to be gradual and follow certain temporal patterns, these patterns must be captured to make accurate predictions. Traditional deep learning models for time series, such as recurrent neural networks (RNN) and their variants, have demonstrated remarkable efficacy in extracting temporal dependencies. Multivariate inputs are mainly dealt with by strategies such as early fusion (raw signal or feature level splicing), mid-term fusion (joint representation in the middle layer of the model), and late fusion (decision level merging).

As the demand for more personalized and adaptive recommendation system grows, the need for real-time computation (Wan, Qi, Xu, Tong, & Gu, 2020), high accuracy, and model interpretability has become a critical concern in the development of such system. Online social media recommendations are often high-density and decisions must be made within strict time constraints, which limits the computational complexity of the underlying models. However, many studies over-pursued accuracy at the expense of efficiency. And understanding the reasoning (Bibal & Frénay, 2016) behind the predictions is the basis of the design of subsequent recommendation algorithms. But they tend to consider only signal variations in a single dimension of time and ignore correlation information between signals. This significantly reduces the interpretability of model decisions.

This research develop a multitasking framework that addresses the dual challenges of real-time pattern recognition and decision interpretability in bio-signal analysis. From a technical perspective, the study's objectives can be distilled into two key points:

- Implement parallel processing of multi-biosignals, facilitating the extraction of complex intersignal relationships while simultaneously enhancing computational efficiency.
- The structural causality of the sequence is preserved to prevent the leakage of future information and to understand the decision process of the model through the a posteriori interpretation module.

Based on the above objectives, we propose CADF (Causality-Aware Dimension Fusion), a novel real-time multi-biosignal modal recognition method. Specifically, we fully consider the spatial and temporal correlations within the original input signal based on an early fusion strategy. Taking into account the interpretability and computational efficiency of the model, the structural causality of the signal is always preserved during the forward inference process, and the feature extraction is accelerated by parallelization of the processing. The main contribution can be summarized as follow:

- Parallel matrix memory unit for long-term dependencies: We propose a parallel matrix storage mechanism integrated with causal convolution to accelerate semantic information processing while maintaining strict structural causality. This design effectively overcomes the limitations of existing methods in capturing long-range dependencies under real-time constraints.
- Enhanced multivariate representation via spatial transformation: A novel spatial mask extraction method is introduced after transformation from one-dimensional to two-dimensional space, to enhance multivariate signal correlation representation. By incorporating sparse 2D causal convolution and adaptive average pooling, the model improves inter-signal communication and robustness, which supports more interpretable signal fusion.
- Lightweight spatiotemporal attention mechanism (STA) for global dependency modeling: We develop a streamlined MultiHead-STA mechanism to capture global temporal-spatial dependencies. This module dynamically computes the relevance between input feature elements and facilitates interpretable and efficient feature fusion across heterogeneous signal components.
- State-of-the-art performance with low complexity: Our model achieves state-of-the-art results across multiple benchmark datasets in bio-signal analysis. Notably, it reduces the number of parameters by half compared to comparable models, ensuring efficiency and suitability for real-time applications.

2. Related work

2.1. Multi-biosignal modal recognition

Human physiological signals are the external manifestations of various physiological activities of the body, which contain rich physiological and psychological information (Huang, Zhan, & Ding, 2025). Typically, multi-biosignal modal recognition can be divided into core tasks such as classification (Lin, Fu, Lin, & Wang, 2022), abnormality detection (Yu, Qiu, Wen, Lin, & Liu, 2016) and assessment (Rissola, Losada, & Crestani, 2021). Utilizing the complementary nature of multiple physiological signals, accurate and comprehensive physiological and psychological state assessment through advanced signal processing and machine learning algorithms provides strong technical support for human-computer interaction, emotional computing and other fields.

Recent efforts have also focused on improving preprocessing techniques and feature extraction from raw biosignals to better support downstream multimodal analysis. For instance, (Zhao, Yang, & Buu, 2025) proposed a structured five-step statistical framework for analyzing heart-rate time-series data, incorporating techniques such as singular spectrum analysis (SSA), matrix profile, and linear mixed models to capture fine-grained physiological variations associated with vaping behavior. Their open-source, modular pipeline provides a useful reference for preprocessing and pattern discovery in biosignal-based behavior modeling.

Biosignal-based biometric systems have also been explored to enhance security in authentication frameworks. A multimodal biometric system (Hussain et al., 2016) combining ECG and heart sound signals is designed for client identification. For a dataset of 20 participants, its recognition accuracy reached 93.92%, demonstrating its robustness. A multi-branch deep learning network incorporating branch attention mechanisms (Mingjin, Zengxi et al., 2023) to dynamically adjust feature importance across biometric modalities, ensuring that noisy signals have less impact on recognition accuracy.

Emotion recognition is another critical application area. An emotion recognition framework (Canento, Fred, Silva, Gamboa, & Lourenço, 2011) integrating multiple biological signals was developed to distinguish between positive and negative emotions, including electromyography, electrocardiogram, and breathing. Further, bilinear fusion of ECG and electrodermal response signals (Singh et al., 2023) has been used to improve the accuracy of emotion classification.

By leveraging the complementary nature of multiple signals, these systems aim to provide more robust and interpretable assessments. However, many traditional systems focus primarily on accuracy, with less attention to real-time capability and decision transparency. Moreover, fusing heterogeneous signal types in a way that preserves temporal structure and improves interpretability remains challenging. Bilinear fusion methods show promise, but few studies address how to maintain structural causality across modalities in real-time systems.

2.2. Deep learning-based feature extraction

Deep learning networks have markedly advanced the field of feature extraction for multivariate temporal series data by automatically identifying inherent patterns and interdependencies, thereby gradually supplanting traditional methods such as statistical analysis and signal processing. This shift is largely attributed to the superior ability of deep learning models to capture hidden temporal dependencies.

Recurrent Neural Networks (RNNs) (Li, Li, Cook, Zhu, & Gao, 2018) were among the earliest models for sequential data, where each output depends on previous inputs, making them suitable for modeling time-based signals. However, they suffer from vanishing gradients. To address this, Long Short-Term Memory (LSTM) (Behera, Jena, Rath, & Misra, 2021) networks introduce gating mechanisms that help retain long-term information. LSTMs and their variants, such as Gated Recurrent Units (GRUs) (Ravanelli, Brakel, Omologo, & Bengio, 2018), have become a standard choice for multivariate time series tasks, including forecasting and anomaly detection. Despite improvements, LSTMs remain computationally intensive and are limited in scalability, especially for high-frequency or long-range data.

Convolutional Neural Networks (CNNs) (Hassan, Hussain, & Qaisar, 2023) have been adopted for extracting local temporal features using the 1D convolution. They are efficient and effective for short-term dependencies. However, CNNs struggle with capturing global context or long-range dependencies, which are crucial for understanding physiological signals over time. Transformers (Xia et al., 2024) represent a major advance. By using self-attention, Transformer models can learn relationships between all time steps simultaneously, without relying on recurrence. Their multi-head attention mechanism captures multiple dependency patterns in parallel. This improves model interpretability and scales better with longer sequences. However, standard Transformers require high computational resources and may not meet real-time constraints without architectural optimization.

In addition to pure Transformer-based models, hybrid approaches that combine CNNs, LSTMs, and attention mechanisms have shown significant improvements. For instance, Zeng, Chen, Jin, Jin, and Du (2022) proposed a hybrid model combining wavelet transforms and LSTMs to enhance the capture of both frequency and temporal dependencies. Temporal Convolutional Networks (TCNs) have also gained attention as an alternative to LSTMs, offering a more parallelizable architecture for long-range dependency modeling while maintaining efficiency (Akbar, Zou, Raza, & Alarfaj, 2024).

In conclusion, deep learning-based methods have revolutionized feature extraction for multivariate time series data, with Transformers and attention mechanisms providing substantial improvements in capturing long-term and inter-variable dependencies. However, balancing computational efficiency with model complexity remains a challenge.

2.3. Feature fusion inference

Feature fusion in multisensory time series data is a critical step in leveraging the complementary information provided by different sensors. Effective fusion strategies enable the model to capture a more comprehensive representation of the underlying processes and improve prediction accuracy. A variety of techniques have been proposed to address the challenges associated with feature fusion, ranging from traditional methods to deep learning-based approaches.

Early approaches (Chaib, Liu, Gu, & Yao, 2017) to feature fusion involved early fusion, where raw features from different sensors are concatenated and input into the model directly. This strategy, while simple, often fails to account for the varying significance of different sensors and may result in inefficient models. To address this, late fusion (Gadzicki, Khamsehashari, & Zetzsche, 2020) has been explored, where individual models are trained on each sensor's data, and their outputs are combined in the final stage. Although effective, late fusion may not fully exploit the interdependencies between sensors.

More recent works have focused on joint learning-based fusion (Bello, Marin, Suh, Zhou, & Lukowicz, 2023) methods that learn sensor relationships during training. Multimodal deep learning models have been widely used, which can simultaneously extract spatial and temporal features from multiple sensor data for better fusion of sensor modalities. For instance, Xu, Tao, Ming, An, and Chen (2020) proposed a model combining CNNs and LSTMs to fuse data from multiple sensors and capture both spatial and temporal dependencies, achieving SOTA performance in sensor fusion tasks.

Another promising approach is attention-based fusion, where self-attention mechanisms are used to weigh the importance of each sensor's contribution (Liu et al., 2021). This method, popularized by Transformers, allows the model to focus on the most relevant sensor features dynamically. Zeng (Chaudhari & Thakkar, 2023) et al. have demonstrated the efficacy of attention mechanisms in improving feature fusion for complex, high-dimensional multisensory time series data.

In summary, feature fusion remains a challenging yet vital aspect of multisensory time series analysis. Recent advancements, particularly through deep learning and attention mechanisms, have significantly enhanced the ability to capture complex sensor interdependencies and improve predictive performance. However, most fusion strategies do not explicitly model causality across modalities, nor do they optimize for inference speed—both crucial for real-world biosignal applications.

2.4. Research gap and motivation

Multivariate time series data, particularly in sensor-based applications, has become increasingly important due to its ability to provide rich and diverse information. However, extracting meaningful features from such data is a challenging task, as it involves capturing both temporal and spatial dependencies across different modalities. Traditional RNNs capture sequential structure but falter with long-term memory and real-time efficiency. CNNs extract local patterns but lack global context. Transformers address dependency modeling but are often computationally expensive.

Moreover, while multi-sensor data fusion methods, such as early and late fusion strategies, have been widely applied, they still fall short in fully exploiting the interdependencies between sensors in a joint learning framework. Early fusion, for instance, can lead to suboptimal representations when the varying importance of different sensors is not considered, and late fusion may not sufficiently integrate the temporal and spatial relationships across sensors. As a result, the predictive performance of these models often suffers, particularly in complex time series forecasting task.

In many existing studies (Abidi, Ienco, Abbes, & Farah, 2023; Sayed, Himeur, & Bensaali, 2023), invisible correlations between features have been found to be better captured by upprojecting 1D time-series data into 2D images and using 2D convolution when processing them (Dang, Raza, Nguyen, Bui-Tien, & Nguyen, 2021). (Zhang, Huang, Zhao, Zhu, & Yu, 2024) further proposed the CENN framework, which integrates a capsule network with multi-head attention for speech emotion recognition based on 2D spectrograms and introduces novel reproducibility metrics such as GLR and CLR. While our approach also adopts 2D transformation and capsule networks, it is specifically optimized for lightweight physiological signal analysis, with an emphasis on computational efficiency and real-time applicability in wearable scenarios.

Dual-dimension features compressed both temporal and spatial information from the signals. The first dimension involves capturing the time-series characteristics of the physiological signals, enabling a clear understanding of temporal dynamics. The second dimension focuses on extracting spatial features, which can be obtained through methods like spatial masking, where the spatial arrangement of the signals is considered to enhance representation. Attention mechanisms that focus on the internal correlations of signals in different dimensions can leverage the inherent information of signals for modal recognition.

The combination of dual-dimension feature extraction and multi-head spatial-temporal attention mechanisms offers a powerful approach for addressing the current challenges in physiological signal analysis. By simultaneously considering both temporal and spatial features and applying causal convolutions to preserve the structural causality of the data, these methods allow for more accurate, robust, and interpretable results.

3. Methodology

Multi-biosignal data can be interpreted in two ways: as a collection of multiple 1D signals or as a single 2D signal where one dimension represents time and the other represents different signal channels (e.g., ECG, EMG, EDA). The primary difference lies in the choice of convolution operation—1D convolution treats each channel independently along the temporal axis, whereas 2D convolution allows the model to jointly learn temporal and inter-signal correlations across channels.

The ability of various modules based on 1D convolution to extract time dependencies has been widely demonstrated. Compared to 1D convolution, 2D convolution has the advantage of capturing joint spatial-temporal features, enhancing the model's ability to learn complex interdependencies between biosignals. This is particularly useful when physiological signals interact or correlate, such as heart rate changes alongside skin conductance. 2D convolution is mostly used in image processing, so to better distinguish the input signals, multi-channel 1D signals are considered as time series data and single channel 2D signals as grayscale images in the subsequent introduction of the algorithm.

As shown in Fig. 1, we propose a dual-branch architecture: a causality-aware temporal encoder and a 2D spatial mask extraction network. This dual representation allows us to leverage the strengths of both 1D and 2D convolutional neural networks. To integrate the features extracted from both branches, we designed a fusion module based on a multi-head Spatial-Temporal Attention (STA) mechanism. This module adaptively weighs the contributions from each branch, enabling the model to focus on the most informative features for accurate classification. In this section we present the model's components in a sequential manner.

3.1. Causality-aware temporal encoder

LSTM effectively manages long-term dependencies through memory cells and three gate mechanisms: forget, input, and output gates. However, LSTM's finite memory capacity may still lead to information loss in extended sequence tasks, primarily due to the gradient vanishing problem (Long, Zhang, & Mao, 2019). To address this limitation, matrix-LSTM introduces an exponential gating mechanism and expands the storage unit from scalars to matrices (Alharthi & Mahmood, 2024). Notably, it incorporates a normalizer state, which aggregates the product of the input gate and all subsequent forget gates, enhancing the model's capacity to retain relevant information over extended sequences.

As shown in Fig. 2, the input to the matrix-lstm cell contains the cell state C_{t-1} , normalizer state n_{t-1} from the last cell, and the current input x_t . Then the current state and normalizer state of the cell can be calculated by Eqs. (1) and (2) respectively.

$$C_t = f_t C_{t-1} + i_t v_t (k_t)^T \quad (1)$$

where T is the transpose operation, f_t represents the forget gate, i_t represents the input gate, k_t and v_t are the keys and values vector. Each gate (input, forget, output) is computed using bilinear operations.

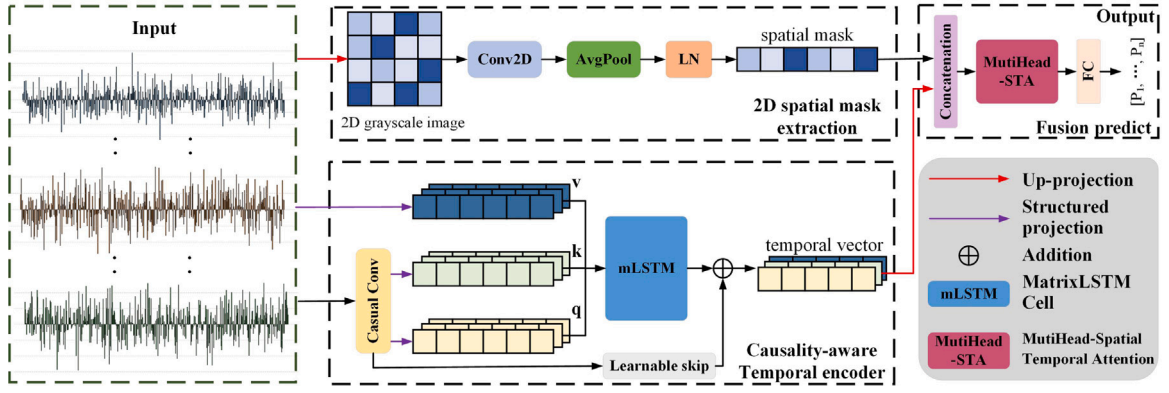


Fig. 1. The overall architecture of CADF. The causality-aware temporal encoder extracts the temporal dependence of the input 1D multivariate time series data, which is merged with the spatial mask extracted on the expanded 2D single-channel data. The fusion prediction module based on the MultiHead-STA mechanism outputs the probability of each category.

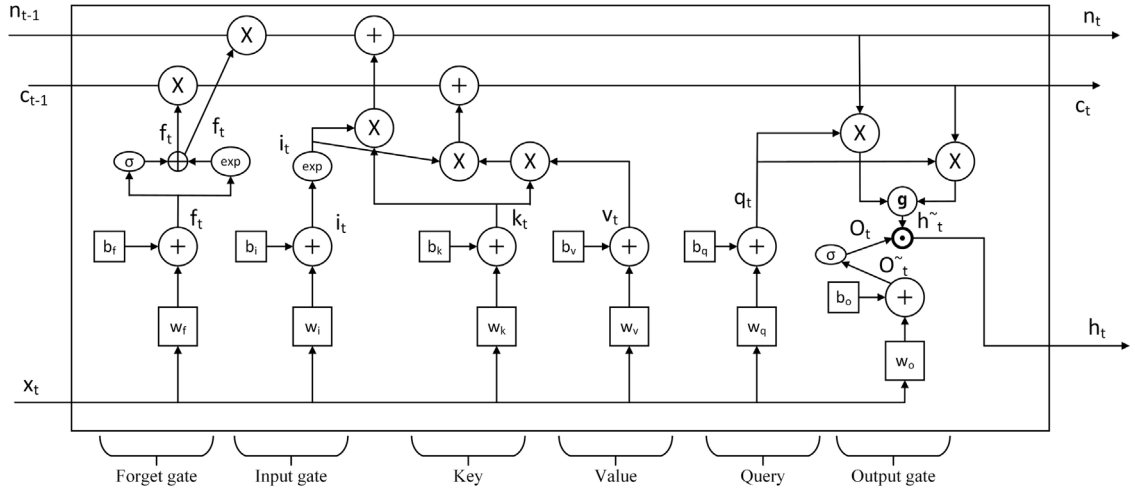


Fig. 2. Schematic diagram of the matrix-LSTM. Cell state, normalizer state and current input were operated by three gates (Forget, Input, Output). Key, value, query vectors are applied to enhance long-term dependency capture.

$$n_t = f_t n_{t-1} + i_t k_t \quad (2)$$

To improve computational efficiency and modify storage decisions, exponential operations are introduced into the original LSTM's forgetting and input gating mechanisms. The forget gate and input gate operations of matrix-LSTM can be represented as Eqs. (3) and (4) respectively.

$$f_t = \sigma(\tilde{f}_t) \text{ OR } \exp(\tilde{f}_t) \quad (3)$$

$$i_t = \exp(\tilde{i}_t) \quad (4)$$

In addition, traditional convolution in time-series modeling often assume symmetric receptive fields, allowing future information to influence current representations. However, in physiological signal processing, such assumptions may introduce unrealistic dependencies and potential information leakage, particularly in real-time or sequential prediction tasks. Our causality-aware temporal encoder addresses this by employing asymmetric, forward-only (causal) convolutions. This approach is theoretically aligned with autoregressive models and causal convolutional architectures, which have demonstrated superior ability to model temporal dependencies in sequential data. To ensure no future information is incorporated into the present prediction, thereby preserving the temporal causality, 1D causal convolution (Tang, Wang, Zhou, Tang, & Lyu, 2024) is applied in calculating the q , k vectors. As shown in Fig. 3 the output at each step t depends only on the current and previous time steps, ensuring the structural causality in

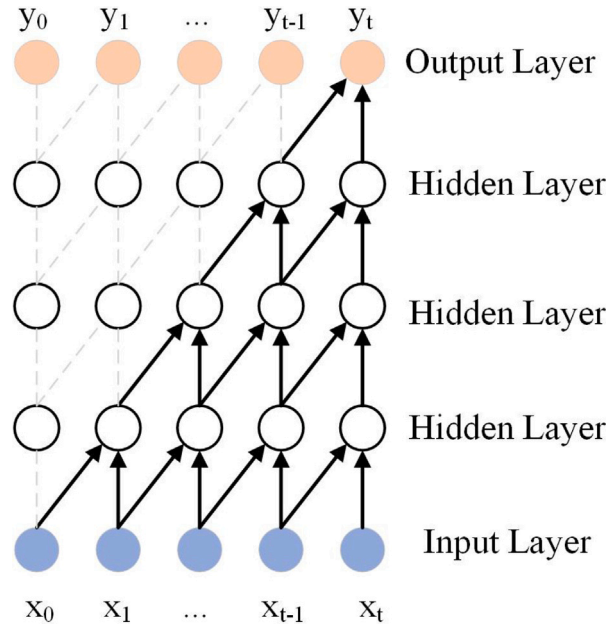


Fig. 3. Schematic illustration of a one-dimensional causal convolution. The input is the time series x_0 to x_t , and the convolution operation is applied only to the current and preceding time steps. The output time series y_0 to y_t is generated, ensuring that the prediction at each time step is based solely on past and present inputs.

Table 1

Algorithm 1 Projection of 1D signals to 2D gray scale images.

Input: One-dimensional multivariate time series signal X

Output: Two-dimensional grayscale image I

1. Receive input X
2. **if** $\text{shape}(X) \neq [\text{batchsize}, \text{len}, \text{num}]$
3. raise error and terminate the process
4. **end if**
5. Check for missing or invalid values in X
6. Unsqueeze X to I with shape $[\text{batchsize}, 1, \text{len}, \text{num}]$
7. Scale I to $[0, 255]$
8. **return** I

the time series. Mathematically, this is represented as Eq. (5).

$$y_t = \sum_{k=0}^{K-1} w_k x_{t-k} \quad (5)$$

where y_t is the output at current step t , x_{t-k} represents the input at time step $t - k$, w_k represents the convolutional filter weights, and K is the kernel size.

The key constraint in causal convolution is that the filter does not look ahead in time, ensuring that the output at time t is computed solely from the current and past values. By enforcing causality in the convolution operation, 1D causal convolution effectively prevent the matrix-LSTM from cheating by using future data, thus ensuring that the predictions are grounded in the correct temporal framework.

3.2. 2D spatial mask extraction

In this research, while the causality-aware temporal encoder learns temporal feature representations with structural causality, the 2D spatial mask extraction branch further captures inter-variable dependencies and ensures robust and scalable feature extraction for complex multivariate signals. Given a multivariate time series signal $X \in \mathbb{R}^{T \times N}$, the first step is to project the data into a 2D matrix $I \in \mathbb{R}^{T \times N \times c}$. The pseudo-code for the projection algorithm is shown in Algorithm 1. This transformation preserves both the temporal structure and inter-variable relationships, making it suitable for 2D convolution operations.

To extract spatial features from the grayscale image, a 2D convolution operation is applied. 1D convolution operation treats input vectors as flattened 1D arrays, while 2D convolution retains the 2D structure of the input and applies transformations that

Table 2

Algorithm 2 Refined model structure search.

Input: Search space S containing candidate Temporal Attention (TA) and Spatial Attention (SA) module configurations, Training dataset D and validation dataset V, Neural architecture evaluation function $E(\cdot)$, Accuracy threshold ϵ .
Output: Optimal architecture A^*

```

1. Define candidate TA and SA module structures in S
2. Sample initial architectures  $A_0, A_1, \dots, A_n$  from S
3. Train a baseline architecture  $Acc_{ref}$ 
4.  $FLOAP_{s_{min}} = FLOAP_{s_{ref}}$ 
5. for  $i = 1, 2, 3, \dots, n$  do
6.   Choose candidate  $A_i$  based on NAS strategy
7.   Train  $A_i$  using D
8.   Compute validation accuracy  $Acc_i = E(A_i, V)$ 
9.   if  $Acc_i \geq Acc_{ref} - \epsilon$ 
10.    Compute  $FLOAP_{s_i}$ 
11.    if  $FLOAP_{s_i} < FLOAP_{s_{min}}$ 
12.       $FLOAP_{s_{min}} = FLOAP_{s_i}$ 
13. end for
14.  $A^* = \operatorname{argmax}_E(A_i, V)$ 
15. return  $A^*$ 

```

exploit spatial relationships within the input matrix. The convolution captures local patterns and correlations among the variables, as well as variable-specific temporal changes. The output of a single convolutional filter is computed as Eq. (6).

$$Z_{i,j} = \sum_{m=0}^{k_h-1} \left(\sum_{n=0}^{k_w-1} W[m,n] I[i+m, j+n] + b \right) \quad (6)$$

Where Z is the feature map, W and b represent the convolutional filter weights and bias term.

After feature extraction, adaptive average pooling technique is used to extract the most relevant features and transform them into a low dimensional space. This operation also adapts to different input sizes to output features of the same scale. The pooling operation for a region R is defined as Eq. (7)

$$W[i,j] = \frac{1}{|R|} \sum_{(m,n) \in R} Z[m,n] \quad (7)$$

Where $|R|$ is the number of elements in the pooling region. The output P (with reduced dimensions N and T) provides a compact representation of the spatial relationships.

To stabilize the features and ensure efficient gradient flow during training, layer normalization is applied to the pooled output. This operation normalizes each feature vector along the spatial dimensions, ensuring that the extracted features are on a consistent scale.

3.3. MultiHead-STA mechanism for fusion prediction

To fuse 2D spatial masks with preliminary features extracted from 1D vectors and enhance model feature representation, this study designed a lightweight MultiHead-STA mechanism to fuse prediction modules. After focusing on the relationship between the variables at each time step, this module captures the dependencies between different time steps. To reduce computational effort and improve the overall timeliness of the module, we refined the redundant structures in the MultiHead-STA mechanism by searching for neural network structures, and the pseudo-code is shown in Algorithm 2. The refined MultiHead-STA architecture is shown in Fig. 4.

The temporal attention module is designed to focus on dependencies across different time steps. It starts with a causal convolution to ensure that only past and present information is considered, maintaining the temporal causality. The output features are then transformed into q , k , and v vectors by linear layers. Then the scaled dot-product attention is computed by Eq. (8), This operation assigns attention scores based on the similarity between q and k , allowing the module to selectively focus on relevant time steps.

$$\text{Attention}(q, k, v) = \operatorname{softmax} \left(\frac{qk^T}{\sqrt{d}} \right) v \quad (8)$$

Residual connections and Layer Normalization are added to stabilize training and improve convergence, which can be expressed as Eq. (9).

$$T_{out} = \operatorname{LayerNorm} \left(\text{Attention}(q, k, v) + \operatorname{CausalConv}(x) \right) \quad (9)$$

The spatial attention module captures inter-variable relationships at each time step, enhancing the model's ability to exploit implicit correlations among variables. The spatial attention weights are computed using a fully connected layer followed by a activation function. To condense the spatially reweighted features, an average operation is applied after feature reweighting. The whole calculation process is shown in Eq. (10).

$$A_{out} = \operatorname{Mean} [T_{out} \cdot \operatorname{softmax}(T_{out} \cdot w_s)] \quad (10)$$

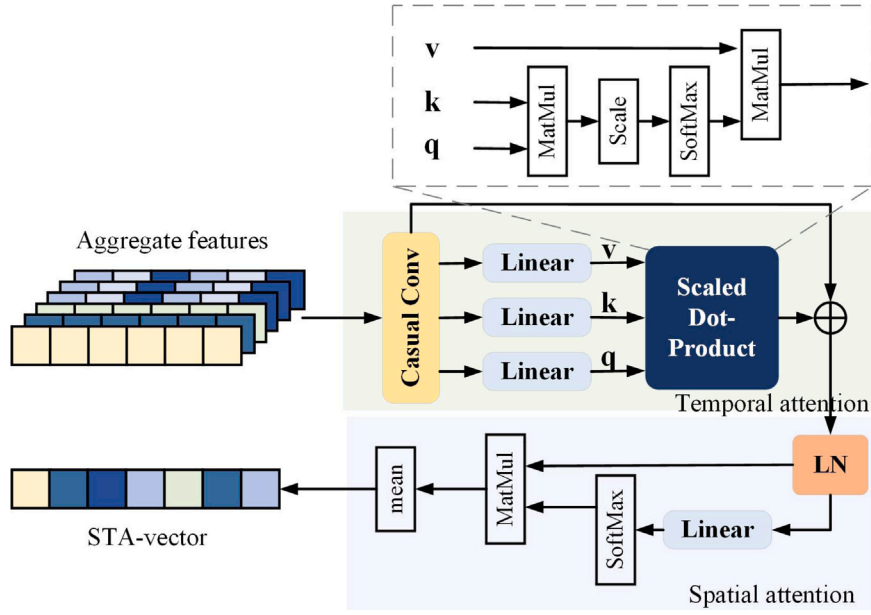


Fig. 4. The architecture of streamlined MultiHead-STA. The input is the aggregated features extracted from the 1D signal and the extended 2D signal, and the output is the STA weight vector.

4. Experiments

4.1. Dataset description and implementation details

To validate the superiority and generalization ability of the proposed method, experiments were conducted on three public datasets: DSADS (Altun, Barshan, & Tunçel, 2010), WESAD (Reiss & Stricker, 2012), and CAP (Terzano et al., 2002).

- The DSADS dataset contains 1,140,000 instances. It includes data from 8 subjects using 9 motion sensors during 19 daily activities. The acquired signals were sampled at a rate of 25 Hz. These signals were then segmented into 5-second intervals, yielding 480 segments for each activity.
- The WESAD dataset contains 63,000,000 instances. It consists of seven common physiological signals collected from 15 subjects in three different emotional states (neutral, stress, and recreation). The duration of the signals varied for each state, and the signals were segmented using a sliding window of length 512.
- The CAP sleep dataset contains 921,700,000 instances. It consists of multichannel polysomnograms recorded during sleep in 16 healthy subjects and 92 patients, and similar to other datasets, these signals were segmented into 512 time windows for analysis.

All three datasets were randomly partitioned into training, validation, and test subsets with a ratio of 7:2:1. To facilitate efficient data loading and reproducibility, the resulting subsets were saved in NumPy binary format (.npy). The proposed method is implemented using PyTorch and tested on a system equipped with an RTX 3090 GPU and 24 GB of RAM. Model parameters are updated through backpropagation with the Adam optimizer, starting with an initial learning rate of $8e-4$. For regularization, a dropout rate of 0.5 is applied within the MultiHead-STA module, along with L2 regularization. The selection of key hyperparameters is discussed in Section 4.6. During training, the cross-entropy loss function enhanced with label smoothing is used, and performance is evaluated using both accuracy and F1-score.

4.2. Comparison results

In this section, we present a comprehensive quantitative comparison of the state-of-the-art of the proposed methods. Three standard metrics are used for assessment: precision (Pre), F1-score (F1), and accuracy (Acc). Precision measures how many of the instances predicted as positive are actually correct. It reflects the proportion of true positive predictions among all positive predictions made by the model. The F1-score is a harmonic mean of precision and recall. It provides a balanced evaluation by considering both false positives and false negatives. Accuracy represents the overall proportion of correctly classified instances, regardless of class.

Table 3

Accuracy comparison experiments were conducted on three datasets including DSADS, WESAD and CAP. The three metrics of classification precision (Pre), accuracy (Acc) and F1-score (F1) are used to evaluate the performance. Bold fonts represent best results.

Methods	DSADS			WESAD			CAP		
	Pre	F1	Acc	Pre	F1	Acc	Pre	F1	Acc
MSCRED	0.7540	0.6428	0.6192	0.8850	0.8471	0.8420	0.6410	0.6081	0.5819
BeatGAN	0.9517	0.7100	0.7818	0.7586	0.6027	0.5172	0.5251	0.5123	0.8437
MNAD	0.5816	0.5799	0.5721	0.7600	0.7254	0.6849	0.7742	0.7613	0.6960
GDN	0.8706	0.8419	0.8251	0.7520	0.6309	0.5590	0.6831	0.6520	0.6569
NERBC	0.8273	0.7975	0.8565	0.7143	0.6507	0.6291	0.7537	0.6147	0.6173
EML	—	—	—	0.9029	0.7884	0.7449	—	—	—
MMFDD	—	—	—	—	—	—	0.6983	0.4919	0.6763
MPHI-Trans	—	—	—	0.8306	0.8162	0.8847	—	—	—
ASK-HAR	0.8931	0.8931	0.8942	—	—	—	—	—	—
Ours	0.9835	0.9821	0.9820	0.9524	0.9525	0.9525	0.7857	0.8095	0.8571

Table 4

Statistical significance results of paired t-tests comparing accuracy with the five baseline methods on the three datasets. P-values were calculated for each of the three metrics of precision, F1 score and accuracy, with bolding under the significant level $\alpha = 0.05$.

Methods	p_{Pre}	p_{F1}	p_{Acc}
MSCRED	0.088	0.087	0.078
BeatGAN	0.140	0.0055	0.218
MNAD	0.215	0.158	0.060
GDN	0.047	0.070	0.075
NERBC	0.141	0.026	0.057

All baseline models were trained and validated using the same data partitions and evaluated using a standardized pipeline. We refer to recent work on general physiological signal classification and anomaly classification tasks, and select the following models that can take into account the classification tasks of three datasets for comparative experiments:

- MSCRED (Zhang et al., 2019): Coding-decoding based classification model for adaptive multi-scale matrix inputs.
- BeatGAN (Zhou, Liu, Hooi, Cheng, & Ye, 2019): General physiological signal abnormal diagnosis model based on adversarial generation.
- MNAD (Park, Noh, & Ham, 2020): A physiological signal analysis method guided by learning and memory.
- GDN (Deng & Hooi, 2021): GNN-based biosignals detection method in multivariate series.
- NERBC (Sun, Portilla, & Otero, 2024): Negative emotion recognition method based on hybrid CNN-LSTM.

To further verify the superiority of the algorithm, we select algorithms that achieve SOTA on a single dataset in the last three years for comparison:

- EML (Vos, Trinh, Sarneyai, & Azghadi, 2023): Focus on monitoring of stress and emotion which reach SOTA in the WESAD dataset.
- MMFDD (Xu, Cong, Chen, & Chen, 2024): Focus on classifying the sleep stages that reach SOTA in the CAP dataset.
- MPHI-Trans (Zhang & Li, 2025): Transformer-based model for multimodal time series data.
- ASK-HAR (Yu & Al-qaness, 2025): performing kernel selection between multiple branches with different RFs, evaluated on the DSADS dataset.

Table 3 summarizes the performance of our method compared to baseline models on the datasets. On the DSADS dataset, our method achieves high performance of more than 98% on all the metrics, which is significantly better than other methods. In particular, compared with the second best method, we improve the mF1 and Acc metrics by about 9%, respectively. For the WESAD dataset, our method also maintains a stable high performance of about 95%, which exceeds the best performing EML method by about 5 percentage points. On the CAP dataset, despite the slight decrease in the overall performance, our method still achieves the best results in the all three metrics, reaching 78.57%, 80.95% and 85.71%, respectively.

In order to assess the performance advantage of the proposed method, we used paired t-test to compare the accuracy metrics of the proposed method with five baseline methods: MSCRED, BeatGAN, MNAD, GDN, and NERBC. The test results are shown in Table 4. In particular, our method shows a significant advantage in the F1 score compared to BeatGA and NERBC. Additionally, the improvement in precision over GDN is marginally significant, with a p -value of 0.0466. These findings provide strong statistical evidence that the proposed approach achieves superior performance, especially in terms of classification balance as measured by the F1 score.

Table 5

Ablation study on the WESAD dataset comparing CADF variants with specific components removed or included. Four evaluation metrics of classification precision (Pre), accuracy (Acc), F1-score (F1), and recall (Rec) were used. Bold fonts represent best results.

Variant	Pre	Rec	F1	Acc
CATE	0.7014	0.7128	0.6909	0.7128
SMEM	0.5513	0.6304	0.5465	0.6304
CATE+ SMEM	0.8014	0.8121	0.8064	0.8053
SMEM+ MultiHead-STA	0.6856	0.6883	0.6677	0.6883
CATE+ MultiHead-STA	0.7923	0.7701	0.7412	0.7701
CATE+ SMEM+ MultiHead-STA	0.9522	0.9512	0.9514	0.9512

Overall, our proposed method demonstrates excellent and stable performance on three different datasets, proving its strong generalization ability and effectiveness. The paired t-test analysis underscores the effectiveness of our approach relative to current baseline models. It is worth pointing out the substantial increase in recall of the proposed method compared to others, which implies a significant reduction in misdiagnosis cases in medical aid diagnosis applications. This result highlights the state-of-the-art of our method in dealing with multivariate time series tasks.

4.3. Ablation study

In this section, we conducted an ablation study on the WESAD dataset to demonstrate the effectiveness of three key components in our proposed CADF model: the 2D spatial mask extraction module (SMEM), the causality-aware temporal encoder (CATE), and the multi-head spatial-temporal attention mechanism (MultiHead-STA). Table 5 reports the performance of different variants of CADF. To ensure fair comparison, we retrained each ablation variant (removing the corresponding module) using the same dataset partitioning and the same hyperparameter settings as the full model. The results shown in Table are the average performance obtained from three runs with three different random seeds.

First, we observe that CATE alone achieves moderate performance with an F1 score of 0.6909 and accuracy of 0.7128. This suggests that capturing temporal causality is critical for characterizing multivariate time series data encoding. The SMEM component in isolation shows lower performance, indicating that spatial information alone is insufficient for optimal results.

Combining CATE and SMEM leads to a significant performance boost, with F1 increasing to 0.8064 and accuracy to 0.8053. This substantial improvement underscores the complementary nature of temporal causality and spatial information in our model.

The addition of the MultiHead-STA mechanism to either SMEM or CATE individually (SMEM + MultiHead-STA and CATE + MultiHead-STA) shows improved performance compared to their standalone counterparts. Notably, CATE + MultiHead-STA achieves higher scores than SMEM + MultiHead-STA, further emphasizing the importance of temporal causality in our approach.

Finally, the full CADF model, incorporating all three components (CATE + SMEM + MultiHead-STA), achieves the best performance across all metrics. This result demonstrates that each component contributes meaningfully to the overall model performance, and their combination yields synergistic benefits.

The ablation study clearly shows that while each component of CADF contributes to its performance, the integration of all three — CATE, SMEM, and multi-head STA — is essential for achieving optimal results on multivariate time series data. The study validates our design choices and highlights the importance of considering both spatial and temporal aspects, as well as their interactions, in developing effective multimodal fusion predict models.

Matrix-LSTM and Multi-STA are also compared with other baseline models and hybrid mechanisms to further illustrate their superiority. As shown in the left bar chart in Fig. 5, matrix-LSTM consistently outperforms RNN, LSTM, and Transformer across all metrics, demonstrating its effectiveness in capturing long-term dependencies. While Transformers perform competitively, RNN and standard LSTM show limitations in modeling physiological time-series data. The right radar chart further validates that MultiHead-STA surpasses other hybrid mechanisms (add, spatial, and temporal) by effectively integrating spatial and temporal attention, achieving superior predictive accuracy. Temporal and spatial mechanisms alone provide partial improvements, while the additive approach proves insufficient. These results emphasize the necessity of jointly modeling spatiotemporal dependencies, with matrix-LSTM and MultiHead-STA offering complementary advantages for physiological signal analysis.

4.4. Explainability analysis

The SHAP-based analysis (Van den Broeck, Lykov, Schleich, & Suci, 2022) provides a comprehensive understanding of the model's decision-making process by quantifying the contribution of each feature. Fig. 6 presents the mean absolute SHAP values, which reflect the average impact of each feature on the model's output. Notably, the EDA signal exhibits the highest influence, followed by temperature (Temp), Acc Y, and Acc X, suggesting their critical role in the prediction task.

To further investigate the distribution and variability of feature contributions, a beeswarm plot (Fig. 7) visualizes the SHAP values for each feature, color-coded according to feature magnitude. The results indicate that Temp and EDA consistently contribute significantly to model predictions, which aligns with the feature importance rankings shown by the mean SHAP value.

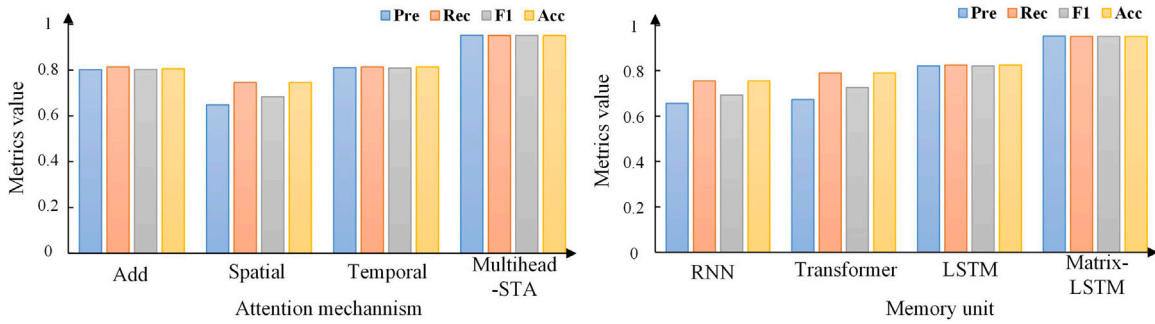


Fig. 5. Performance of core composition compared to existing methods on the WESAD dataset. The right is a comparison between matrix-LSTM and the other three baseline models, and the left is a comparison between MultiHead-STA with the other three hybrid mechanisms.

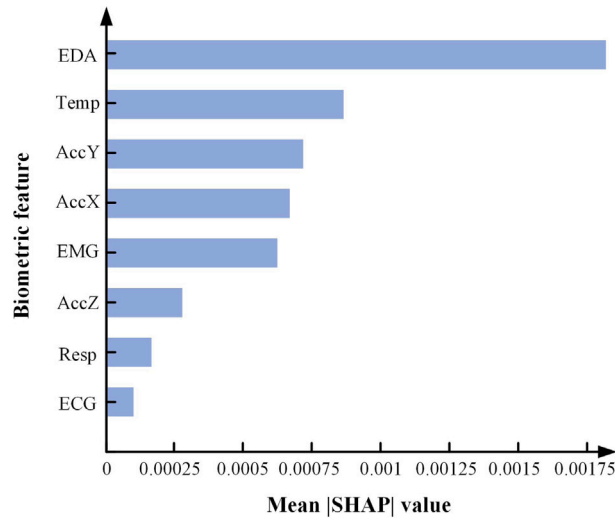


Fig. 6. Importance ranking of biometric features based on the SHAP value. The contribution of different biometrics features—including electrodermal activity (EDA), electrocardiogram (ECG), electromyogram (EMG), temperature (TEMP), respiration rate (Resp), and three-axis acceleration (AccX, AccY, AccZ) to the model's predictions is calculated. Higher SHAP values indicate greater influence on the output, reflecting the relative importance of each input modality in the task.

A deeper analysis reveals that lower temperatures have a limited impact on the model's predictions, suggesting that body temperature alone may not be a reliable indicator of a subject's mood in these cases. In contrast, higher body temperatures tend to exert a more pronounced positive or negative effect on model decisions, implying a non-linear relationship between temperature and the target variable. Additionally, EDA signals, irrespective of their specific eigenvalues, consistently influence model predictions. This finding aligns with human physiological responses, as EDA is closely associated with emotional arousal, thereby reinforcing the model's interpretability from a causal perspective. Consequently, these results suggest that the model exhibits a certain degree of causal reasoning in its decision-making process.

4.5. Convergence, time and space complexity

Fig. 8 illustrates the convergence characteristics of our model during training. The loss curves demonstrate rapid and stable convergence, with the training loss (blue curve) exhibiting a sharp decline in the initial epochs, indicative of efficient learning and optimization. The validation loss (orange curve) follows a similar trajectory, stabilizing after approximately 20 epochs. This behavior suggests robust generalization and absence of overfitting, while also highlighting the model's ability to efficiently locate the optimal solution without requiring excessive training iterations.

Table 6 presents a comparative analysis of the computational complexity of our method against several benchmark approaches. The analysis focuses on three key metrics: the number of parameters (Params) and floating-point operations (FLOPs). Our method demonstrates superior efficiency and performance, achieving the highest accuracy while utilizing significantly fewer parameters and FLOPs compared to all other approaches. This represents a substantial improvement over competitors such as MSCRED, which achieves slightly lower accuracy but requires more than four times the parameters and three times the FLOPs.

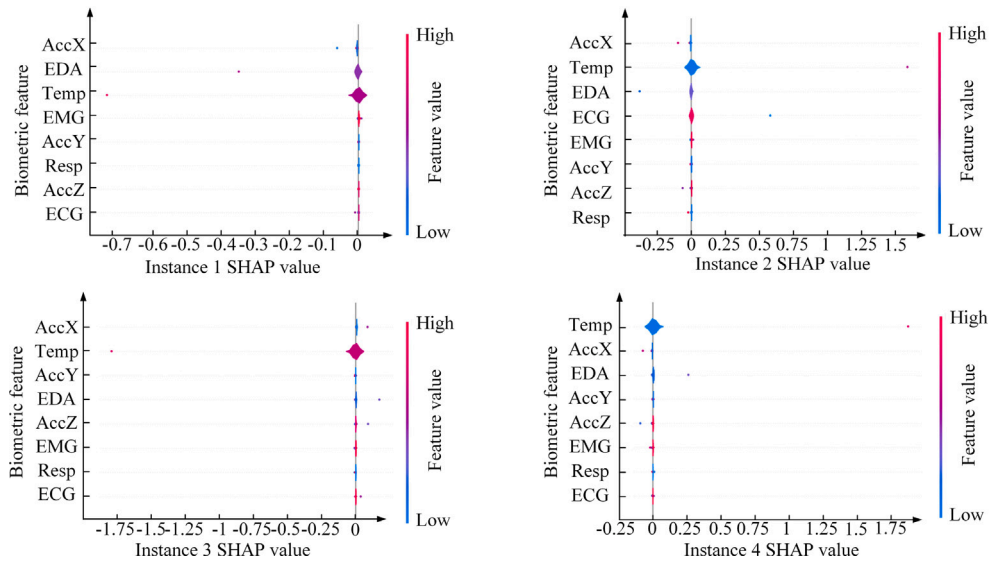


Fig. 7. Beeswarm plot of SHAP values for 4 samples. Positive SHAP values indicate a positive contribution to model prediction, while negative values indicate an inhibiting effect.

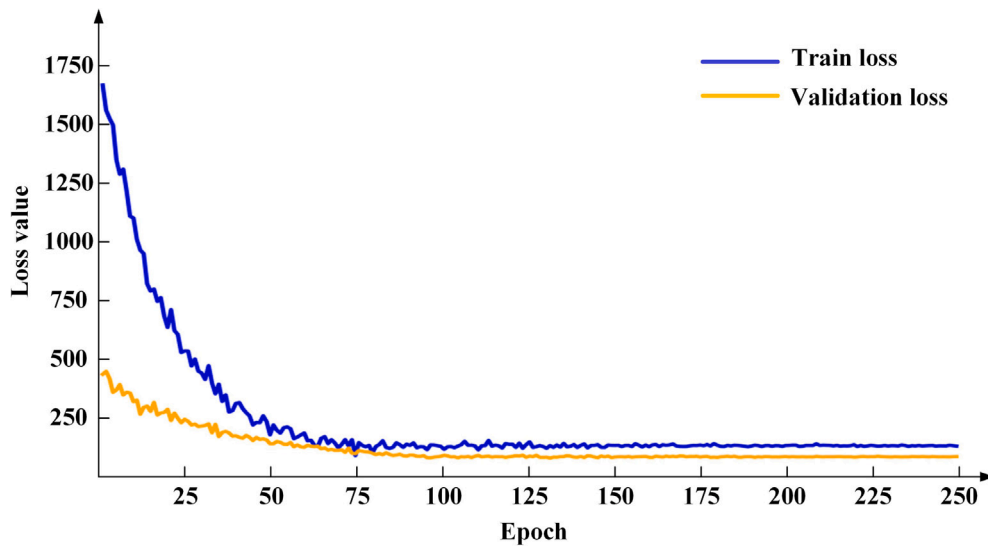


Fig. 8. Visualization of model loss over the training process. Blue line indicate the training loss recorded over a total of 250 epochs, while orange line represent the corresponding validation loss.

Table 6

Comparison of the model parameters and computational effort. The number of parameters (Params), floating point operations per second (FLOPs) and inference time (InT) are reported for each model under identical hardware settings and a batch size of 1. Bold fonts represent best results.

Method	Acc	Params (M)	FLOPs (M)	InT (ms)
MNAD	0.6849	0.0296	14.7875	4.51
MSCRED	0.8420	0.0281	13.8125	24.37
NERBC	0.6191	0.0148	7.3152	8.23
Ours	0.9512	0.0059	4.1404	2.59

Table 7

Impact of different hidden dimensions and attention head numbers on model performance. This table compares the effect of varying these key architectural parameters on both classification performance and computational cost. Evaluation metrics include precision (Pre), accuracy (Acc), F1-score (F1), and recall (Rec), along with model complexity indicators: number of parameters (Params) and floating-point operations (FLOPs). Bold fonts represent best results.

Dim	Head	Params (M)	FLOPs (M)	Pre	Rec	F1	Acc
16	1	0.0018	1.5021	0.8324	0.8345	0.8343	0.8345
	2	0.0019	1.5497	0.9084	0.9093	0.9079	0.9093
	4	0.0020	1.5989	0.9373	0.9372	0.9370	0.9372
32	1	0.0056	4.0704	0.9432	0.9434	0.9431	0.9433
	2	0.0057	4.0912	0.9524	0.9525	0.9525	0.9525
	4	0.0059	4.1404	0.9522	0.9512	0.9514	0.9512
64	1	0.0195	12.2841	0.9376	0.9344	0.9342	0.9346
	2	0.0196	12.3199	0.9425	0.9414	0.9414	0.9414
	4	0.0197	12.3691	0.9440	0.9445	0.9439	0.9445
128	1	0.0718	41.3526	0.9353	0.9366	0.9357	0.9369
	2	0.0719	41.3602	0.9285	0.9280	0.9262	0.9280
	4	0.0720	41.4094	0.9470	0.9460	0.9461	0.9459

The efficiency of our method is particularly evident when compared to MNAD and MSCRED, which not only have lower accuracy but also significantly higher computational demands. NERBC, while having fewer parameters and FLOPs than some other methods, falls short in terms of accuracy. Given a sample length of 512 and a sensor sampling frequency ranging from 25–60 Hz, the data acquisition time for a single sample is a minimum of 8 s. Considering the proposed method exhibits an inference time of only 2 ms, it comfortably satisfies the necessary real-time processing constraints. The proposed approach decisively addresses the typical trade-off between computational efficiency and performance by achieving the highest accuracy with the lowest computational complexity. This compact model architecture significantly reduces memory requirements in both the training and deployment phases, improving the applicability of our method in real-time applications and resource-limited environments.

4.6. Parameter sensitivity analysis

To assess the impact of different model structure hyperparameters, output dimensions (dim) and attention heads (heads), on the prediction performance and model complexity, we conducted a parameter sensitivity analysis. Table 7 summarized the experimental results.

From the results, we observe that increasing the dimension (dim) generally improves model performance in terms of all metric up to a certain point, beyond which the performance starts to decline. Specifically, at dim number of 32, the model achieves the highest performance, with an F1-score of 0.9525 when using 2 heads and 0.9514 when using 4 heads. This suggests that a moderate latent dimension provides a good balance between representation capacity and generalization ability.

However, when dim is further increased to 64 and 128, we notice a slight degradation in performance. For instance, at dim number of 128, the model's F1-score drops to 0.9262 (2 heads) and 0.9461 (4 heads). This can be attributed to overfitting, where an excessively large feature space reduces the model's ability to generalize effectively to unseen data. Regarding the number of attention heads, we observe that increasing heads from 2 to 4 leads to minor improvements in performance across most settings, particularly for dim number of 16 and 128.

In addition, we also analyze the case when the number of heads is reduced to 1, which simplifies the attention mechanism. Interestingly, we find that using a single attention head has little impact on parameter size and computational cost, making it an attractive option for resource-constrained settings. However, its impact on performance varies with dimensionality.

When dim is small, reducing the number of heads to 1 leads to a noticeable drop in precision, indicating that multiple heads are important for capturing diverse feature subspaces under limited representation capacity. Conversely, when dim is large, the performance gap between 1-head and multi-head configurations becomes smaller, suggesting that the increased feature dimensionality can partially compensate for the loss of attention diversity.

From the parameter and FLOPs analysis, we observe that increasing dim leads to a significant rise in computational cost. When dim is increased from 16 to 128, the number of parameters grows from 0.0019M (2 heads) to 0.0719M (2 heads), while FLOPs increase from 1.5497M to 41.3602M. This demonstrates the quadratic complexity of the model concerning the feature dimension, highlighting a trade-off between model efficiency and accuracy.

Similarly, increasing the number of attention heads from 2 to 4 results in a slight increase in both parameters and FLOPs, but the impact is relatively minor compared to changes in dim. For example, at the dim number of 128, increasing the number of heads from 2 to 4 raises the FLOPs from 41.3602M to 41.4094M, an insignificant increase compared to the gains in predictive performance. This suggests that while adding more heads slightly improves performance, it does not substantially increase computational cost, making it a viable option for enhancing model robustness.

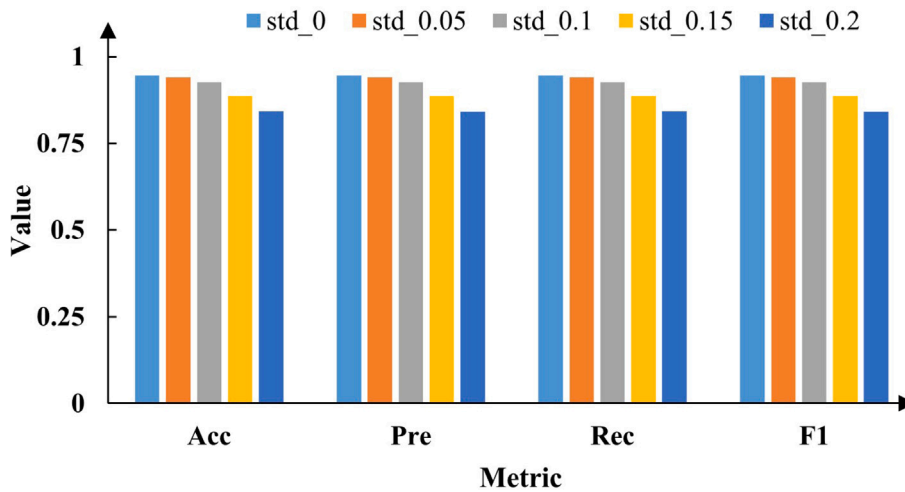


Fig. 9. The robustness of the proposed algorithm to noisy data. Gaussian noise was injected in the test set with a standard deviation(std) between 0 and 0.2.

4.7. Robustness to noisy data

Biosignal data are inherently susceptible to noise contamination caused by environmental disturbances and variations in sensor placement or wearing conditions. To assess the robustness of the proposed algorithm under such realistic conditions, we conducted an experiment using the WESAD dataset in which Gaussian noise was injected into the test set at varying levels of intensity. Specifically, zero-mean Gaussian noise with standard deviations ranging from 0 to 0.2 was added to simulate different noise conditions.

The performance of the model under each noise level was evaluated using four standard classification metrics: accuracy (Acc), precision (Pre), recall (Rec), and F1-score (F1). The results are summarized in Fig. 9. As the noise level increased, a gradual decline in performance was observed. However, the model maintained relatively high performance even under the highest noise level tested (standard deviation = 0.2), achieving an accuracy of 84.36% and an F1-score of 84.21%. These results demonstrate that the proposed algorithm exhibits strong robustness to noise in biosignals, which is critical for real-world deployment where signal quality may be inconsistent.

5. Discussion

With the rapid development of artificial intelligence and wearable devices, physiological signal modality recognition has emerged as a prominent research hotspot. However, notable research gaps remain. Most existing studies emphasize improving algorithmic accuracy while overlooking computational efficiency. This presents major limitations in real-world applications and partly explains the slow adoption of machine learning in this domain. For instance, commonly used models such as CNN- or RNN-based architectures perform well in terms of accuracy but often struggle with high-frequency, multimodal signals due to their large model sizes and long inference times—making them unsuitable for deployment on resource-constrained devices like wearables (Zhao et al., 2020) (Chen, Wang, & Zhang, 2019). In contrast, the causality-aware dimension fusion (CADF) proposed in this study presents significant theoretical and practical contributions that address these limitations.

Firstly, CADF maintains the structural causality of input signals, effectively preventing future information leakage—a critical issue often neglected in conventional models. Unlike transformer-based approaches that model global dependencies at the cost of temporal integrity, CADF enforces causal order to preserve the logical sequence of events. SHAP-based interpretability analysis further reveals that CADF incorporates a level of causal reasoning in its decision-making process. This is a vital contribution to the field, addressing the existing gap in explainability and causality-aware modeling (Lundberg & Lee, 2017).

Secondly, CADF integrates interpretability techniques that provide deeper insights into model decisions. Compared to traditional “black-box” models widely used in healthcare, CADF enables feature-level transparency by visualizing key temporal segments and their contribution to predictions. This enhances trust and usability in critical settings such as remote monitoring and clinical diagnostics (Chen et al., 2020).

Moreover, CADF adopts a lightweight architecture that drastically reduces the number of parameters compared to deep learning baselines. Experiments show that CADF maintains high prediction accuracy while achieving significant reductions in model size and inference time, making it highly suitable for edge computing and embedded applications. This minimizes dependence on cloud infrastructure and facilitates integration into real-time systems (Wang, Liu, & Huang, 2021).

When compared to the broader literature, CADF stands out through several key innovations: a causality-aware temporal encoder that mitigates issues like vanishing gradients while preserving sequence structure; a flexible spatial mask extraction mechanism that improves feature selection across modalities; and an advanced multi-head self-attention design that enhances fusion and

generalization. These architectural advances have demonstrated superior accuracy across cross-validation tasks, establishing CADF as a robust and efficient alternative to existing methods.

6. Conclusion

This paper introduces CADF, a novel framework for real-time multi-biosignal modal recognition. CADF effectively addresses the challenges of capturing long-term dependencies and inter-variable correlations by integrating three key innovations: (1) a causality-aware temporal encoder with parallelized matrix memory units and exponential gating to enhance temporal feature extraction while preserving causality, (2) two-dimensional spatial mask extraction module to model multivariate correlations, and (3) a refined MultiHead-STA mechanism to facilitate the fusion of temporal and spatial features.

Extensive experiments on DSADS, WESAD, and CAP datasets demonstrate the superiority and generalization ability of CADF. Our method consistently outperforms existing approaches, achieving up to 98% accuracy and delivering significant improvements across multiple evaluation metrics. Ablation studies confirm the necessity of each component, with the full CADF model exhibiting the best overall performance. Furthermore, efficiency analyses reveal that CADF achieves a favorable trade-off between computational complexity and accuracy, demonstrating its suitability for real-time physiological signal analysis.

Despite these advances, some limitations remain. Current models in physiological signal modality recognition primarily rely on post-hoc interpretability methods (e.g., SHAP, LIME) or causal convolution structures to preserve the temporal causality of signals. However, such approaches often fail to achieve a balance between deep interpretability and model performance. Post-hoc explanations can be misleading, inconsistent, or computationally intensive, and may not accurately reflect the model's internal reasoning (Hooshyar & Yang, 2024; Lemanczyk, Bartoszewicz, & Renard, 2024). Moreover, models that employ causal convolutional mechanisms are typically constrained to shallow interpretability structures, making it difficult to provide transparent reasoning in high-stakes domains like healthcare (Sokol & Vogt, 2023). Future research will be organized around the following expanded directions:

- **Unified multi-modal integration:** We will develop a fully integrated framework that fuses data from multiple sensing modalities — such as ECG, EEG, and PPG — in a cohesive manner. This will allow the model to perform joint reasoning across sensor streams, enabling more accurate and context-aware predictions.
- **Causality-aware intrinsic interpretability:** We aim to move beyond traditional post-hoc methods by embedding interpretability directly into the model architecture. Inspired by recent advances in inherently interpretable neural networks (Ditz, Reuter, & Pfeifer, 2023), the goal is to build models that offer transparent and verifiable insights without requiring auxiliary explanation modules.

In conclusion, CADF represents a significant advancement in multi-biosignal modal recognition, offering improved accuracy, computational efficiency, and potential for interpretability. These contributions have a wide range of applications in many fields such as emotion recognition.

CRedit authorship contribution statement

Jing Tao: Writing – original draft, Methodology, Conceptualization. **Zhuang Li:** Writing – review & editing, Software. **Lin Wang:** Writing – review & editing, Supervision. **Dahua Shou:** Supervision, Funding acquisition.

Ethical considerations and privacy discussion

All three datasets used in this study are publicly available benchmark datasets that have been released with appropriate ethical approvals and informed consent at the time of their original publication. Each dataset includes a disclaimer stating that the data may be used for scientific, non-commercial purposes, provided that proper credit is given to the original data providers. We have strictly adhered to these terms by appropriately citing and acknowledging the datasets in this work. Moreover, the datasets used contain only anonymized physiological signals without any personally identifiable information (PII), ensuring that the privacy of participants is not compromised. Our proposed method focuses solely on analyzing depersonalized physiological patterns and does not involve re-identification of individuals.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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